

Funding the Commons - Impact Certificates and Impact certificates and Impact Markets



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Join Owen Cotton-Barratt for his talk titled, "Impact Certificates and Impact certificates and Impact Markets," part of Protocol Labs' "Funding the Commons" virtual summit for individuals and organizations building new models of sustainable public goods funding and value alignment in open source networks. For more information go to: <https://fundingthecommons.io/>

[Music] [Applause] [Music] my next speaker owen cotton-barrett is a

0:07

research associate in the future of humanity institute at the university of oxford focusing on decisions in the face

0:14

of great uncertainty he's an effective altruist director of the research at the center for effective altruism and holder

0:21

of a d field in mathematics owen will be speaking to us about the about impact certificates and impact

0:27

markets the benefit of markets and altruistic impact and what's needed to achieve them

0:34

welcome owen

0:41

uh hi i'm really happy to be here um

0:48

that screen check okay yes

0:54

great yeah um so i i'm gonna talk about uh impact certificates and impact

0:59

markets i think that many of the original uh ideas here originally traced back to uh paul christiano and dennis

1:05

drescher's recently been helping me uh work out some things so this is a kind of

1:11

relatively uh theory talk about how we could be reconceptualizing

1:16

uh impact and how um the doing of good things uh we relate

1:21

to that and what types of things even can be tradable um but i i think there might be a lot of value on the table by

1:28

getting a good system set up here so what what are we even talking about uh roughly i'm thinking of impact is

1:38

managing to change in the supply of a public or maybe private good um uh

1:44

there's a couple of modifiers here uh if you ask change what does it that's supposed to be change relative to

1:51

the idea is that there's some kind of counterfactual world history where you didn't take a particular action
um

1:57

mostly i think that we can not bother too much with this and we just have this intuitive notion of like oh you know

2:03

what what do we mean by by impact and uh that'll flow through okay but i think

2:08

it's interesting to uh be able to bottom out what we mean uh

2:15

uh if we need to and then i also think that i'm regarding it as the responsibility for that change

2:21

and that uh the key notion is that um you like do something good and then the

2:26

the impact flows back to the person responsible um so that's the basic setup

2:33

uh and then maybe the kind of like an interesting question is well if this is a thing that belongs to people often for things that

2:39

belong to people we assign people property rights over them um and they can say oh well you know i want to give

2:45

this away i want to assign this to to somebody else and then you can start building markets on top of

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that um and we can ask the question for impact as we've defined it here should we be able to uh uh give assignation

2:59

rights and have people say actually uh this impact which was coming to me i am passing on uh to another party

3:06

um and if we do then that could be the first step towards allowing markets in

3:11

impact um uh like impact is this slightly rarefied

3:18

uh abstract notion and i think that there are several possible benefits of having markets and

3:26

they're kind of the usual benefits but transferred over to a new context i think that several of these benefits we

3:32

can get through uh mechanisms which don't make any reference to

3:39

uh uh like rarefied impact like i think that i've put here is the first benefit

3:44

exposed evaluation of a charitable work or other other work

3:50

that's aiming uh to be good um and prizes are a normal way of doing that

3:55

but i i think that uh the reconceptualization can help uh to spot places where we can

4:02

uh port over other market mechanisms and and there can be something useful about it

4:07

so yeah the normal story for why x post evaluation is better is

4:12

that uh you you have more information like we you just get to see more oh did

4:18

that turn out well um uh versus if you're uh restricted to funding things

4:23

uh beforehand then you have to go on the basis of do people have a plausible

4:28

story and private information is

4:34

like bites harder in that case uh i think that

4:39

yeah so this can allow for better integration of diffused information where um

4:46

somewhere before kind of the final price for some set uh the place allow whichever individuals

4:54

have pertinent information as to uh the eventual value of something to

5:00

get that information into the market uh by buying or selling and

5:06

[Music] the markets can be a good way of integrating that information

5:13

and then i think another potentially big deal is that it can incentivize excellence if you have a

5:20

system where uh markets can give big rewards to

5:26

people who do things that are really good then it incentivizes people to look for hey how could we do something that's

5:31

great um and i think that uh uh just humans often incentivized by uh

5:40

uh things like uh getting recognition but also getting rich and uh there's a lot

5:45

of energy that goes into um oh how can we make a for-profit company that's that's gonna uh uh make

5:52

us rich and i think that if we could get that um same innovation and energy going

5:57

into uh how can we do things where a for-profit model doesn't work but we still want to be really

6:03

ambitious and aim for uh big changes in the world then uh this is something where markets and

6:09

impact can be pretty helpful um so the basic i

6:14

like conceit uh which i'm gonna illustrate through to a metaphor here and so the

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idea is like rather than just accepting the impact from a good action that you've done and and you in this case

6:26

could be a person or it could be an organization uh or maybe it can be some other kind of uh

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entity but you you uh like intercept with a bottle um and you say no i'm not

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taking the impact for this instead i'm going to put it into storage and now it's a transferable unit which i can go

6:43

and sell um and yeah so you can think of this like an nft for a particular good deed that has

6:51

been done um and the metaphor with the bottles uh the key

6:58

step here is that owners can later open the bottle um uh kind of un uh

7:05

meaning that now whoever uh gets to uh drink it or whatever uh gets the

7:13

impact that was originally bottled and so um if you have tokens then at this point

7:18

they would become untradeable they would record uh who

7:24

uh was attributed the uh associated impact

7:31

um and maybe if people often think of nfts as like a market in fine art this might be

7:38

sort of more like a market in fine wine uh ultimately uh the value of things is

7:44

propped up by people wanting to uh consume them at some future point

7:50

um so that's the basic idea and then there's lots of reasons why

7:56

this is pretty hard to get set up uh one is that impact attribution is often

8:01

implicit and it's complicated you have a thing which a bunch of people have collaborated on and it's just not clear

8:08

who gets which amount of credit um and and it's a mess to try and sort out and that means

8:13

that it's a mess for anybody to try and uh sell the associated impact

8:19

because the buyers don't really know what they're getting um i also think that right now we don't

8:26

have good ways of recording anything there um

8:33

and like the recording thing is we can deal

8:38

with and then finally you know nobody cares we don't have people who are saying oh yeah i really want to buy into

8:44

uh these impact markets there isn't the demand for it um

8:51

so one other concept is i want to draw a distinction between impact certificates which are the things i was representing by bottles tokens uh denoting the

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rights to attribute the impact associated with some particular

9:04

specified actions and impact shares which entitle uh the

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owners to share a future stream of impact uh uh certificates from whatever

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their own entity is uh i mean impact shares are just kind of shares uh but in in the impact context i just wanted to

9:21

draw this distinction because i think that sometimes when i've heard people uh talking about ideas here uh

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the two can be blurred and i think that conceptually they're distinct conceptually they're both useful and i

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expect that if we ever do get to the point where we're building mature ecosystems around these we will be using both

9:42

so what would be needed to set up impact markets i think there's uh maybe three central

9:49

things there's uh bookkeeping uh we want some way of uh

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recording okay who owns the impact associated with particular uh actions um

10:01

uh such that people can uh trust uh however that bookkeeping process is done

10:07

um i i think that uh kind of connected to the bookkeeping is you want some basic way to validate

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that the the records uh of the things that are being written down line up with real world impact that

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they are uh supposed to represent um uh you don't want somebody just claiming hey i i did this amazing thing

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um uh when uh and and getting paid for it when uh

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uh it's kind of if essentially just fortunate um i think that the second is needing

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social buying you need buy-in from lots of different parties you want buy-in from whichever bodies are potentially

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doing the uh good deeds which would be bottled and issued as certificates um uh you need

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buy-in from uh marketplaces to facilitate

11:00

uh the uh uh purchase of these um uh but i

11:05

think and and maybe buying from

11:10

if you want to get benefits of speculators and markets who maybe want the buy-ins uh from

11:16

uh speculators but i think that the key thing is from the ultimate buyers uh those people who are uh have an interest

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in eventually consuming um uh the bottled impact and so

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here i might think of uh this as

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substituting in for when people currently give money to uh charities or

11:39

other uh good uh causes um and saying well hey maybe they would

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prefer to buy uh impact if you're giving money to a charity it i

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think that the social contract is such that implicitly you are buying some impact from the charity you're saying well i

11:58

uh want to give this money because i trust that it will be used for good things um

12:05

and this is saying well there's a different setup where you could uh still

12:10

uh buy the good deeds but you get to find out exactly what you've bought uh rather than doing it uh

12:15

all on trust that something uh good will happen and also if these things get implemented

12:22

legally i think that the point of consumption of these certificate

12:28

should be tax deductible uh there's a bunch of questions of how you can actually set

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that up but that's the stage which is a very tightly analogous to making

12:39

donations um and then there's a bunch of infrastructure uh that one would want to

12:46

have in a mature ecosystem uh helping to support this

12:52

i think maybe the

12:58

most key elements here the ones three certificate auditors you want um

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some bodies which do the work of investigating

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um certificates to check out and say hey yeah this is legit the thing which it is

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claiming happened really did happen um because if that is left to each individual buyer to

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do uh then that presents a massive amount of overhead on the market and uh

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is likely to uh make uh just yeah purchases

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unwilling to enter and maybe you also early on want to have uh

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certificate issuers which help to facilitate that process it's also possible that you could set it up in

13:49

such a way uh that people's self-issue you might want to combine uh three and four here you might want to have some uh

13:57

relatively trusted bodies which arise as uh hey we will issue certificates and uh

14:04

uh we will develop a reputation which people get trust in that we do a good job of

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auditing the things that we issue certificates for and therefore any certificate issued by

14:16

us can be trusted um but that could also uh just live as a layer on uh

14:22

on top of this i'm imagining that a mature market would have competition between multiple different

14:28

auditors and uh trying to audit things efficiently without spending too much

14:33

resources but also to a high enough standard that it establishes a high degree of trust

14:40

and then there's a question of okay well if we would eventually need all of these things what is the critical path uh

14:46

towards setting up markets here um and of the three bookkeeping social buying and infrastructure i really think

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that social buy-in is uh uh the key aspect uh we want people

14:59

using this and then that will the more that there's demand and people using it the more that will drive

15:05

everything else uh uh being built to meet that demand

15:12

but i think that um uh bookkeeping and uh some aspects of infrastructure are

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helpful for facilitating uh that social buy-in if there's a clean implementation

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if people feel like oh yeah okay uh it it's it's very obvious what's happening and i feel trust in uh the technical way

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that's been set this has been set up then that can facilitate the social buy-in but i don't think that it's the only part of social buying which matters

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um i outlined earlier why

15:43

i am excited about impact markets what i think that the the big upsides here are i also think that there are

15:52

uh a number of potential downsides and

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i'm just gonna highlight a handful which seem to me like the biggest issues

16:04

um one is uh that at least on the a naive implementation of this

16:10

uh impact markets could incentivize likely harmful charities and the reason is that

16:16

there's an asymmetry where if a charity uh does a bunch of work i mean sorry i'm

16:23

i'm saying charity but it could just be uh actions uh which

16:28

uh somebody is doing where they're hoping to make

16:34

uh money by selling the impact um yeah if somebody does something that's

16:39

very risky and that there's uh some chance of getting uh extremely bad

16:45

outcomes and some chance of getting extremely good outcomes um

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the value of the associated certificates is generally capped below by zero

16:56

um it's due to know zero is what you get if nobody in the market is willing to pay for this um

17:02

and this could mean that uh if there's a high tail upside that

17:07

people are incentivized to take those actions uh even if the

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expected value of the thing is strongly negative

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now this isn't a problem which is unique to impact markets uh in general it's a

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question of negative externalities and there's a question of how society should coordinate to uh

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try to tax negative externalities or otherwise uh incentivize people to uh not create

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them um but i do think that it can bite the issue can buy it particularly cleanly

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here and so it's worth as some extra attention yeah it's worth pointing out that

17:52

um you might think well uh markets generally let people send a price signal

17:59

uh that they they think something is overvalued by shorting uh a stock uh but

18:05

because the eventual resolution here can't go below zero that doesn't provide a direct way for um

18:13

markets to say that they think that something is uh actually strongly net negative uh so this could be this could be worked

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on um and uh maybe there are some uh [Music]

18:25

interesting innovative uh solutions which could work here this

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could be regarded as a paul christiani pointed this out to me part of the general problem

18:36

uh that impact markets uh as we're currently conceiving them

18:42

don't solve the problem of moral trade where you uh

18:49

meaning that you fail to reach pareto superior outcomes uh which everybody

18:54

would um uh prefer uh because they're nobody's kind of first preferred option

19:00

uh by trade um i think that there's also issues

19:06

about changing the incentives uh otherwise on um uh people pursuing good work and it

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might push people towards things which are easier to uh

19:20

turn into a impact that can be sold um or easier to

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uh or to put more work into marketing and explaining why this thing that you

19:32

did in the past was really good of course this is to some extent a duel of issues which already come up for

19:38

charities where they can be incentivized to pursue work that they think

19:44

uh will uh uh sell well and they'll be able to persuade people to give them money to do rather than for having done

19:52

um and uh to put a lot of money into fundraising for um uh

19:58

uh uh their future work so there are traps there it's not

20:03

um i don't have a particular reason to think that they are larger than the current issues that uh

20:10

could be worth for the investigation and in general there's a kind of like hesitance of well

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this could be a large shake up in how

20:22

uh charitable work is funded and uh some people have been thinking about

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that some but uh we definitely haven't worked through all of the implications of that and maybe there are unknown effects of how it affects uh of

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the impacts on on the ecosystem uh of moving to this system so i'm

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this one makes me particularly interested in uh experiments which have some ability

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to say okay no let's let's close up shop uh this uh seems like it's not going uh in

20:54

a in a productive direction um uh but there are complications of how that

21:00

interacts with incentivizing trust from people okay and so then in the last part of the

21:06

talk i want to talk about what uh we're calling virici 0.1 um

21:12

the numbering system and name is to to suggest that this really is uh kind of

21:17

early stage thinking um and the idea here is that if we want functional markets and we

21:23

want people uh having a lot of trust in those markets i think that that uh depends on people having a good

21:31

understanding of how the markets work um

21:37

and if i go and i try to buy a stock on an

21:42

exchange i don't need to have gone and looked into all of the legal contracts that are supporting the particular um uh

21:51

type of stock that i am buying and to check and i i can just have like some

21:57

generalized trust in the exchange and uh

22:02

uh that the thing is regulated in some way which uh uh s

22:09

stops them selling uh 50 times as much stalkers actually exists

22:16

etc um and i think that there are other ways well if we just set

22:23

set up common expectations of how things are going to work that could be pretty helpful and so

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this is my current best guess uh for the social norms uh that we should try to

22:37

set up as the rules of markets uh impact markets if we were doing experiments

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today to establish impact markets and

22:49

uh you know i'm sure there'll be mistakes in here but i really think it is worth trying it if we're making attempts to

22:55

start with our best guesses and so there are a few different properties the first is verifiability that impact has

23:02

occurred uh uh i think that it is helpful if there is gatekeeping around the market

23:09

uh which makes it clear to market entrance that uh things which are on sale there have

23:16

passed some level of auditing uh that they really happened um i think that it's also really

23:24

important to have clear ownership of uh rights and not to try and sell impact in

23:30

cases where there's too much ambiguity over ownership rights i think that we want some kind of

23:36

consumption mechanism uh this equivalent of uh unbottling uh

23:42

of the impact treating it like fine wine rather than fine art um and i think that we want to

23:50

set things up in such a way that they are incremental and that this is

23:56

acceptable to existing uh interests

24:02

in in the markets um and so i'm not gonna say a bit more about each of these four

24:09

uh and yeah they they don't yet address most of the risks i mentioned before um

24:14

uh these are more uh rules where i think creating common knowledge around this uh would help give

24:22

the best shot of uh creating healthy

24:27

markets which a lot of people are actually happy buying into um

24:33

yeah i mean the basic issue with verifiability is that it's hard to facilitate trust and buyers if people

24:39

can't verify whether the impact has happened um and

24:45

uh as a matter of ui and making the process

24:51

relatively seamless for buyers i think it's helpful if there is some relatively trusted zone where they can operate in

24:58

and they can say well now i'm just going to look at what the things are claiming rather than thinking anytime i need to make a

25:03

purchase here i i need to uh go and do a massive amount of uh

25:10

due diligence and there are various different ways that this could be set up it could be um

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that one has a marketplace and one accredits

25:22

various different auditors and then uh buyers coming in can do filters on i

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only want to see things which have been approved by uh this kind of auditor

25:34

um yeah so the issue of rights i think is

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interesting i think that there are lots of cases where uh say i work on a project with someone

25:48

um and then we say and then then we're like oh well that did well and now i want to go and sell the

25:55

impact uh for my work on this this can create adversarial dynamics uh and

26:00

incentives between the people who were the collaborators because now they're each incentivized to produce a public

26:06

narrative uh for why their contributions were particularly important and other people's contributions uh were

26:13

relatively unimportant and i think that that could be um pretty unhealthy and destructive

26:19

and so i think that we uh want to avoid having that and

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often i think that the way to do this will be via explicit agreements among collaborators

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about uh how impact will be shared um similar to the advice for

26:39

uh uh startup founders to uh agree a share of equity very early on

26:47

uh in in the process so um society has to grapple that with this

26:53

uh uh in in various cases and there are um a bunch of situations where uh ownership

26:59

already ends up uh uh pretty clear and like society has allocated a standard

27:04

answer to this if somebody owns a company uh 100 then presumably they own uh

27:11

all of the uh impact certificates that that company is generating um similarly for copyright

27:17

and if there is uh impact associated uh uh uh coming from

27:24

the uh the copyrighted material um by default i think that should go to the

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copyright owner um i think that one can specify a clear explicit

27:38

divisions uh and in general if you're doing something by yourself uh it's uh pretty simple

27:44

relatively unclear cases include donating to a charity and so

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uh a challenge for getting impact markets off the ground is that i think that

27:55

right now charities often couldn't uh in good faith issue

28:02

uh certificates of impact because there is significant ambiguity as to how much

28:08

the rights for the good deeds that they have done lies with the charity rather than the donors to the charity if they

28:14

have some kind of um uh explicit expectation setting with the donors uh uh then i think we'll clear that up

28:23

um i think uh you get a similar thing with

28:28

um people who are employed by a charity where uh i think that there's an

28:34

implicit social contract that uh they

28:40

are doing that in part because they regard their work as contributing to the good

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that the charity is doing uh but i think it's ambiguous as to how big their contribution is

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and just lots of other activities where there's a collaboration uh but there's no been no explicit discussion about

28:58

divisional credit um

29:04

then consumption uh i was pointing to earlier i think that this has a couple of different benefits

29:09

one is that it helps to provide social proof um that there are ultimate buyers

29:14

uh interested uh and these buyers are about removing tokens from circulation

29:19

and so i think that this should uh support trust in

29:25

underlying value of uh tokens because it builds up a track record um

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and less than fears of bubbles i also think that it helps with uh the the final property of being incremental

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because it provides continuity with the current practice um in the current practice

29:44

uh implicitly when somebody is doing something good they're just consuming uh

29:49

the impact immediately and this is saying hey there's another operation

29:54

that they could take uh where they say no we're going to hold off on that um and bottle it and put it into a

30:00

certificate and then maybe that can be sold but if they don't do that uh uh we're just in the status quo one

30:07

um and i think that that continuity is helpful because if you imagined uh

30:12

something without a property like this then there could be uh a bunch of people saying hey does this

30:19

impact markets thing mean that now lots of charities need to be paid again for

30:24

the work they did didn't we already give them money for that uh ahead of time and i think that that

30:31

could create bad feeling and could uh make

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parties uh who like the parties would be giving money here uh uh somewhat

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disinclined to engage with the process for fear of being exploited

30:50

um yeah so this is the the basic principle behind

30:55

incremental i think that at every stage we want to uh

31:00

set things up so there's just incentives for uh people to uh come in and start using

31:07

this uh and that nobody is directly losing out as a

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result yeah in order to avoid creating uh kind of entrenched opposition uh uh

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to the system i think that uh this can be totally

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i think this is totally achievable um and i think that uh it's normally the good pathway to to go for if uh you

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want a new economic technology to to to take over um okay so uh i'll wrap up that uh just

31:41

to as as and then we can take a few questions and the

31:46

final comment is this yeah i kind of hopeful that impact markets uh

31:52

could be a big thing and could transform um how a lot of analysis is done and a lot

31:59

of incentives in the ecosystem of people trying to do good things and i'm really interested to to

32:07

see how they develop and i i think that this kind of conference is great because the people

32:13

at conferences like there so exactly the right people to be going and building experiments

32:23

i think so and for a comprehensive look at markets and altruistic impact that acknowledges the risks as well as the

32:30

upsides and we have some time for questions and i know there are a few in the chat if you don't mind i'll surface

32:35

them here the first question is from actually another one of our

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forthcoming presenters andrew kritsch and he asks oh and i know you've thought about chapter value

32:46

to reward counterfactual um impacts like making a good attempt that falls short of someone else's

32:54

i think having people engage in public discourse about chaplain value attributions could really help reward

32:59

people who almost made it and thereby incentivize more attempts to do good

33:04

could you comment on whether something like that seems good or what challenges it would raise

33:13

yeah okay great question um

33:22

i think that the main challenge uh uh

33:27

that i'm worried about is this thing that i alluded to about uh trying to sell impact when

33:33

uh uh there's unclear ownership of the rights i think that it could create

33:39

adversarial uh incentives for people to uh overstate their contributions and

33:44

understate uh those of others um

33:53

however i i see i see like the prize that you're pointing at and i i think maybe there could be something good

34:00

there i feel more like oh this is worth more thought not i know

34:07

how to make this work um i'm in the moment mildly pessimistic about

34:13

uh getting it to work in a straightforward way but there are lots of

34:18

uh more complex implementations which might dodge uh

34:25

many or more all of the issues

34:31

thank you and i'll i'll ask another question this one's from one oh and do you think bottling and

34:37

consuming is required to make impact certificates work why can't consumption be a time bandwidth oriented i.e the current

34:44

holder of the certificate like with art markets

34:50

yeah so i do think that there i i think that this is

34:57

a a pretty significant bifurcation in how you want to conceive of and set up impact markets i

35:04

think that the uh world where you say you treat it like the art markets and

35:11

you say uh the current holder is deriving value makes some sense but it has

35:17

it loses significant amounts of cleanness relative to the uh one where it uh which has consumption and it i

35:24

think it may lose significant amounts of the benefit so some of the cleanness that it loses is

35:31

uh you can no longer just look at something and say oh how good was uh this as as a way of informing uh how

35:38

much you should care about uh buying it you sort of want to look at well how much of the value has already

35:43

depreciated um and uh

35:52

you're going to be baking in some expectations of well how much

35:58

uh society uh seems to care about this thing but it

36:04

reduces the ability to say well we want to to use this as a kind of like consequentialist uh a mechanism um

36:13

i think that it uh also probably makes it significantly harder

36:20

to set up in a way that gives you anything like tax deductibility um uh uh for buying these things i think

36:27

that uh uh for my my understandings of like the principles

36:34

of how tax deductibility is supposed to work uh not anchored on uh the laws of any any one country the idea is when you

36:42

give money to a thing for which you do not receive like material benefit that's the point at which you should be given

36:47

the tax deduction and i think that uh the fine art model of uh impact

36:54

certificates never has such a moment um and therefore it is hard to uh

37:01

uh work out how the tax deduction should work i think that the uh version with consumption does have such a moment and

37:08

that's better and then uh i also think that the

37:14

fine art model makes it harder to uh

37:20

incentivize profit motivated uh speculators and analysts to enter into

37:26

the market um because uh you ex uh in this model you expect by default

37:34

uh most certificates to depreciate with time maybe there'll be some which go up in value because

37:42

people just realize how great the thing was so you can still get some of this but essentially there will be a significant tax

37:48

on anyone who is uh trying to enter into this uh from the perspective of well uh

37:54

i'm not gonna be the person ultimately like buying these and wanting to hold the impact but i can add good

38:00

information to the system and i i can earn a profit for doing that and so i'll i'll come and uh help things out and and

38:07

those people are just a bit more pushed out of the system which means that i expect to get uh

38:13

uh less of the benefits of relatively efficient markets uh out of that all of

38:18

that said you totally can do a version uh without consumption um

38:24

uh uh there's also the uh the issue of maybe you're more likely to

38:30

just get bubbles um uh in in the version without consumption uh because there's nothing uh uh propping up the

38:37

uh uh uh the price of things um but the yeah the original proposal

38:43

for certificates of impact uh uh by paul christiano and catcher grace uh didn't have consumption uh

38:49

i i feel less optimistic about it doing really good things but maybe i'm wrong

38:58

thank you for ending on a very humble note there i think we have time for one more question just a quick one from evan miazono

39:04

oh and do you know of anyone building the market you described are you planning to and what support would be most helpful

39:12

um i am personally not planning to but i

39:18

know uh uh uh dennis uh dresher who

39:24

uh uh helps me with some of the thinking

39:30

about the uh the vuichi standards has been uh thinking about

39:35

what would be needed to get a fairly minimal uh implementation of this uh set up on

39:43

top of uh nfts uh uh in some existing uh

39:49

chains uh i yeah i am actually i haven't touched base

39:57

with him recently enough to know what the current status of that is i think maybe we were thinking oh we should publish the blog

40:03

post on virici and we haven't quite got that

40:08

great well we would love to keep uh apprised of this research and there's some excitement in the chat about this

40:14

idea and um now it's time so i'd like to thank owen again for his time

40:22

[Music] [Applause] [Music] you